



Public Bus Stop Accessibility and Service Gaps Across Chennai

Exploratory analysis of a bus stop audit

210 stops audited | 85 variables

Problem, Objectives and Method



Exploratory analysis, not causal inference

Objectives

- Pedestrian and wheelchair movement
- Bus-to-kerb boarding conditions
- Shelter roof, shade, seating, capacity
- Encroachment, drainage and safety
- Stop name and route information
- Land-use context and priorities

Method

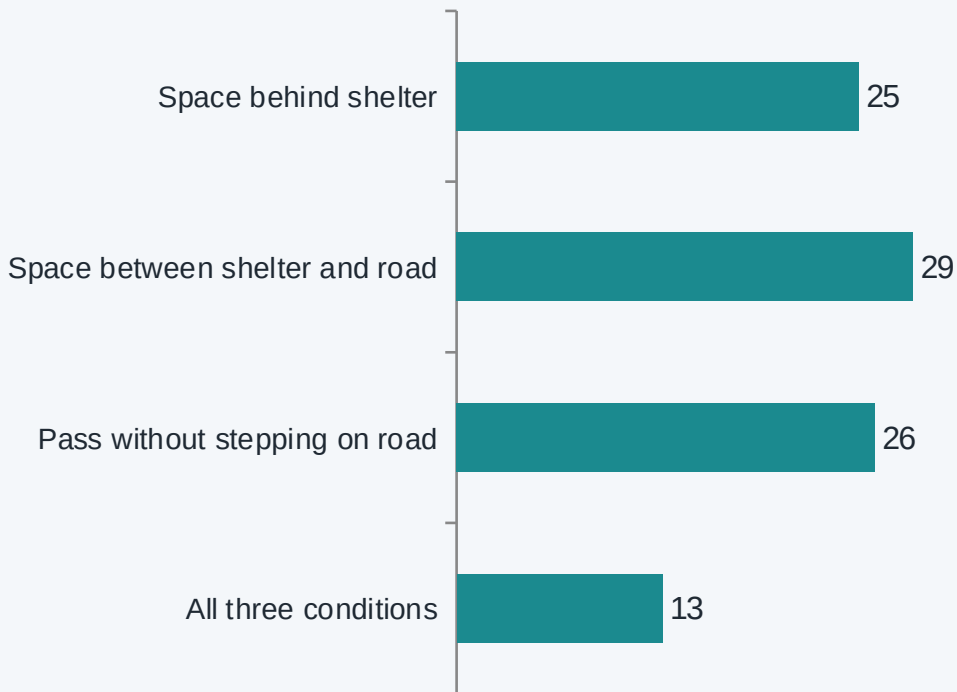
Data loading, quality checks, whitespace and date cleanup, duplicate removal, missing-value review, derived indicators, frequency analysis and charts.

Pedestrian access score

Counts Yes for: space behind shelter, space between shelter and road, continuous footpath movement (0 to 3).

Pedestrian Access Is Weak

Share of audited stops meeting each condition (%)



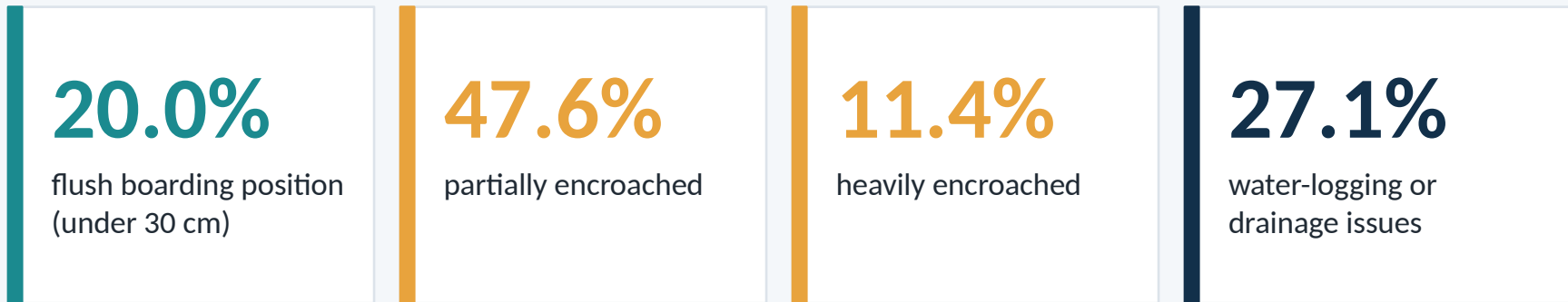
12.9%

of stops satisfy all three
pedestrian-access conditions

Fewer than 3 in 10 stops allow
pedestrians or wheelchair users to
move freely.

Boarding, Encroachment and Drainage

Physical conditions around the stop



Four in five stops have a boarding gap of 30 cm or more, and about 59% are encroached to some degree.

Lighting and Passenger Information

Safety and information gaps

17.6%

inadequate or no lighting

34.8%

no route information displayed

27.6%

no clearly displayed stop name

Over a third of stops give passengers no route information at all.

Recommendations

Priority actions for a targeted improvement program

Pedestrian access

Footpath continuity, clear movement space, accessible approaches

Boarding gaps

Better kerb and platform design; align bus stopping positions

Encroachment

Manage parked vehicles, vendors and construction material

Lighting

Upgrade lighting at unsafe or poorly lit stops

Information

Clear, current route info and legible stop names

Drainage

Fix water-logging around affected stops

Shelters

Review capacity, shade, seating and roof condition

Limitations and Conclusion

Limitations

- Audit sample may not cover every Chennai stop
- Some fields are missing
- Responses are partly subjective
- Results should not be generalized city-wide

Conclusion

Gaps are concentrated in pedestrian movement, boarding distance, encroachment, lighting and passenger information. The derived indicators can help prioritize stops for infrastructure and service upgrades.